

Name:	ID
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A. Differentiate between Shell molding and Shell Casting:

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B. Differentiate between vacuum molding and vacuum Casting:

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C. Differentiate between Cold Box molding and Hot Box molding:

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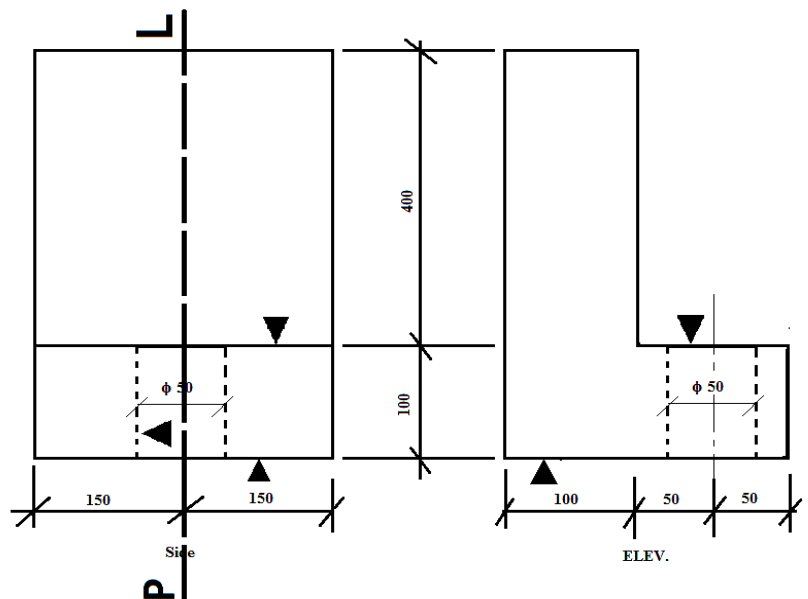
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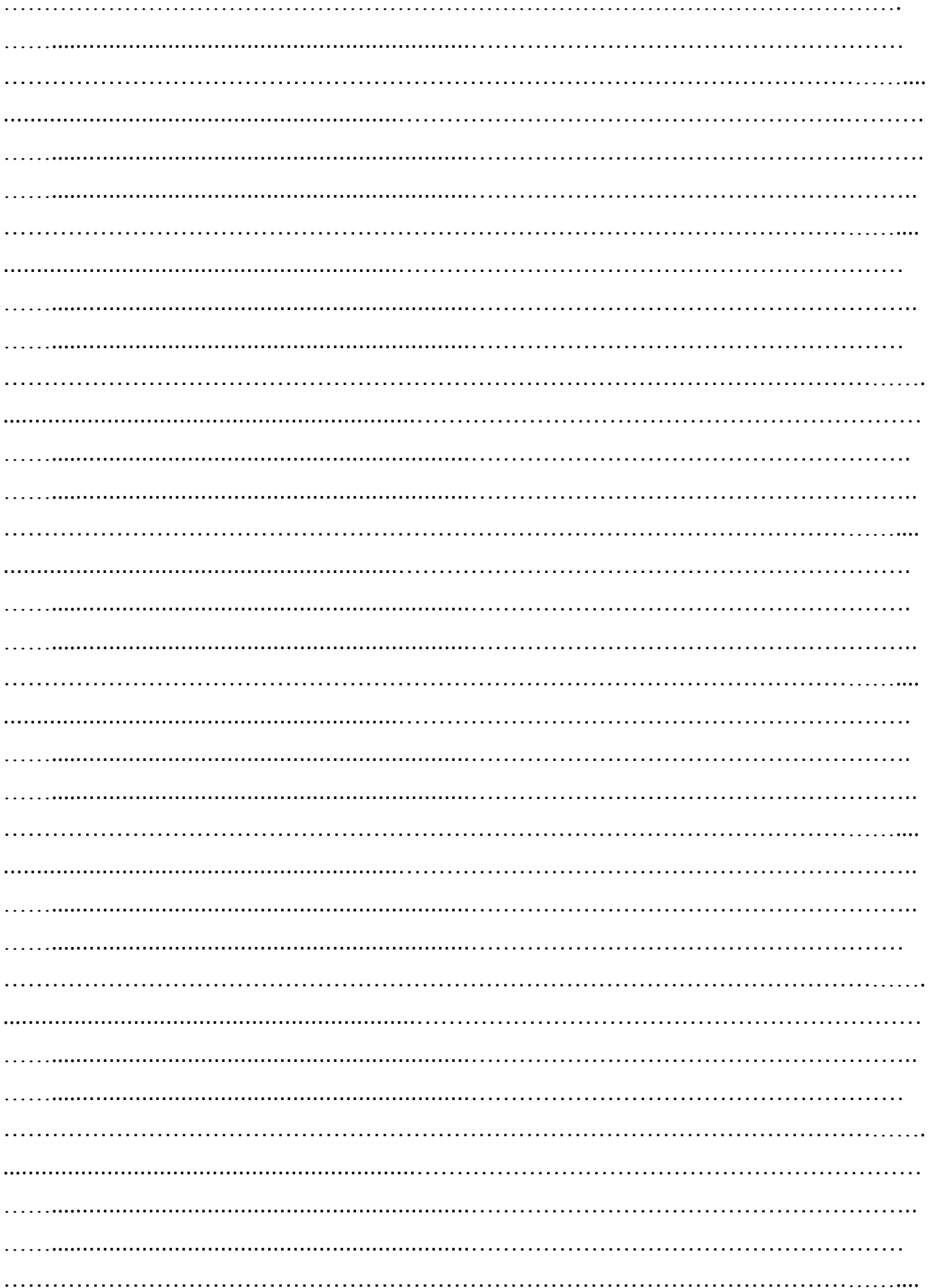
D. It is required to sand cast the shown product from gray cast iron (7.2 g/cm^3) using the drawn PL, knowing that the product will be machined after casting on the surfaces with (▼) mark only. Find:

1. Suggest suitable casting modifications,
2. Suitable flask dimensions.
3. Core dimensions.
4. Casting pattern dimensions.
5. Volume and weight of molten Metal.
6. The forces acting on mold.

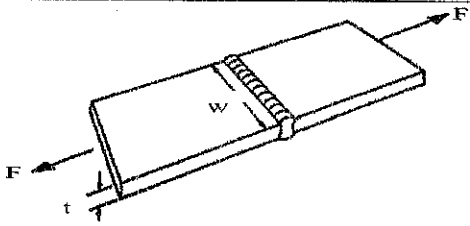
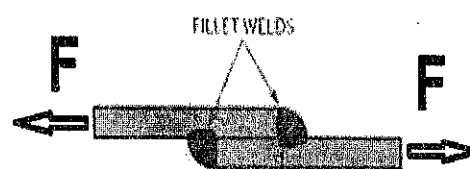
Take: Shrinkage allowance = 3 mm/m

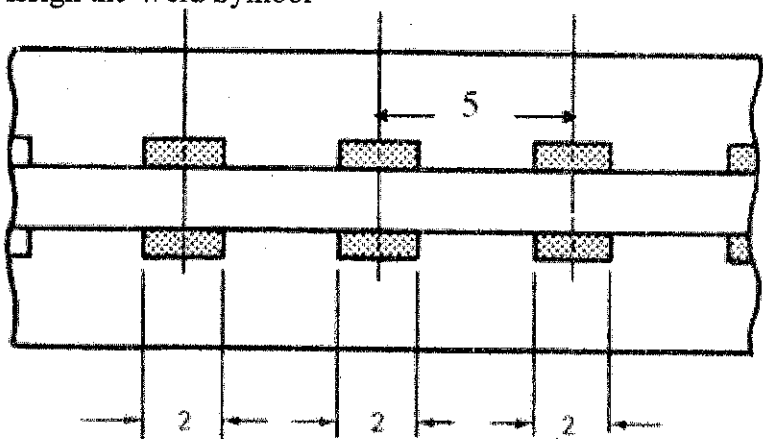
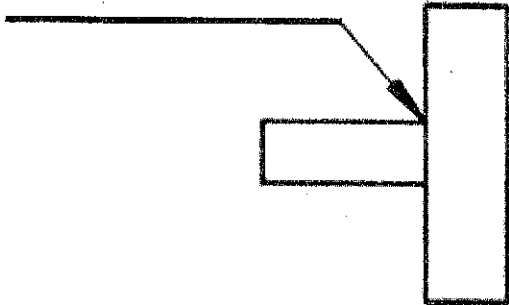
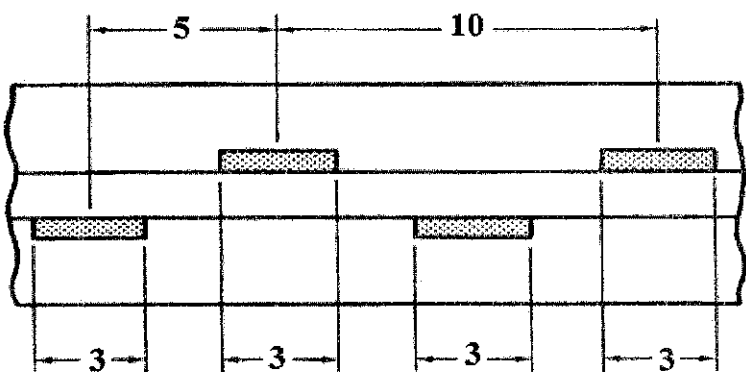
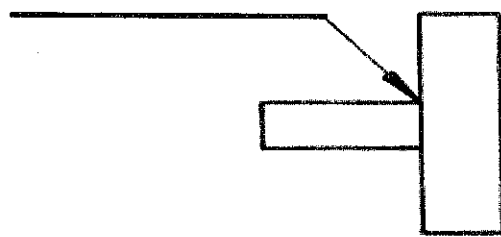
Machining allowance = 2 mm





SEC.	Code.	أسم الطالب / (بالغة العربية ويخط واضح)
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Calculate [F] the force needed to break the Joint, Two plates, Steel (St.52), [t = 5 mm & W = 20 mm] SMAW Electrode E7018 Note that: [1 Ksi = 6.89 MPa] ($\tau = 0.6 \sigma$)		2 Marks
Calculate [F] the force needed to break the Joint, Two plates, Steel (St.52), [t = 5 mm & W = 20 mm] SMAW Electrode E7018 Note that: [1 Ksi = 6.89 MPa] ($\tau = 0.6 \sigma$)		2 Marks

Assign the Weld Symbol 	 Weld Size 3/8 in	2 Marks
Assign the Weld Symbol 	 Weld size 1/4 in	2 Marks

Define The WPS	2 Marks
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Assessment Methods / LO Matrix

Assessment Method	LO1	LO2	LO3	LO4	LO5
Formative Assessment Methods	■	■	■	■	
Summative Assessment Methods			■	■	■